

Tutorial 1

$$1.4.) \quad I = \frac{Q}{t}$$

$$I = \frac{300}{30}$$

$$\underline{I = 10 \text{ A}}$$

$$1.6.) * I = \frac{dq}{dt}$$

$$a.) \quad t = 1 \text{ ms} \Rightarrow I = \frac{30 - 0}{2 - 0} = \underline{15 \text{ A}}$$

$$b.) \quad t = 6 \text{ ms} \Rightarrow I = \frac{30 - 30}{6 - 2} = \underline{0 \text{ A}}$$

$$c.) \quad t = 10 \text{ ms} \Rightarrow I = \frac{0 - 30}{12 - 8} = \underline{7,5 \text{ A}}$$

$$\begin{aligned} 1.10.) \quad I &= 10 \text{ kA} \\ t &= 15 \mu\text{s} \end{aligned} \Rightarrow Q = I t$$
$$= 10(10^3) \times 15(10^{-6})$$
$$= \underline{0,15 \text{ C}}$$

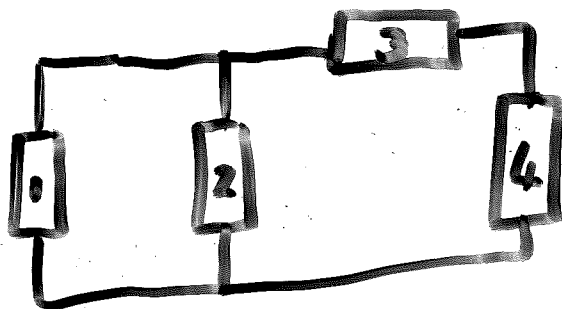
$$1.11.) * I = \underline{90 \text{ mA}}$$

$$t = 12 \text{ h} = 12(60)(60) = \underline{43200 \text{ s}}$$

$$\begin{aligned} \textcircled{1} \quad Q &= It \\ &= 90(10^{-3}) \times (43200) \\ &= 3888 \text{ C} \\ &= \underline{3.89 \text{ kC}} \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad E &= VQ \\ &= (1.5)(3888) \\ &= \underline{5.83 \text{ kJ}} \end{aligned}$$

1.17.)



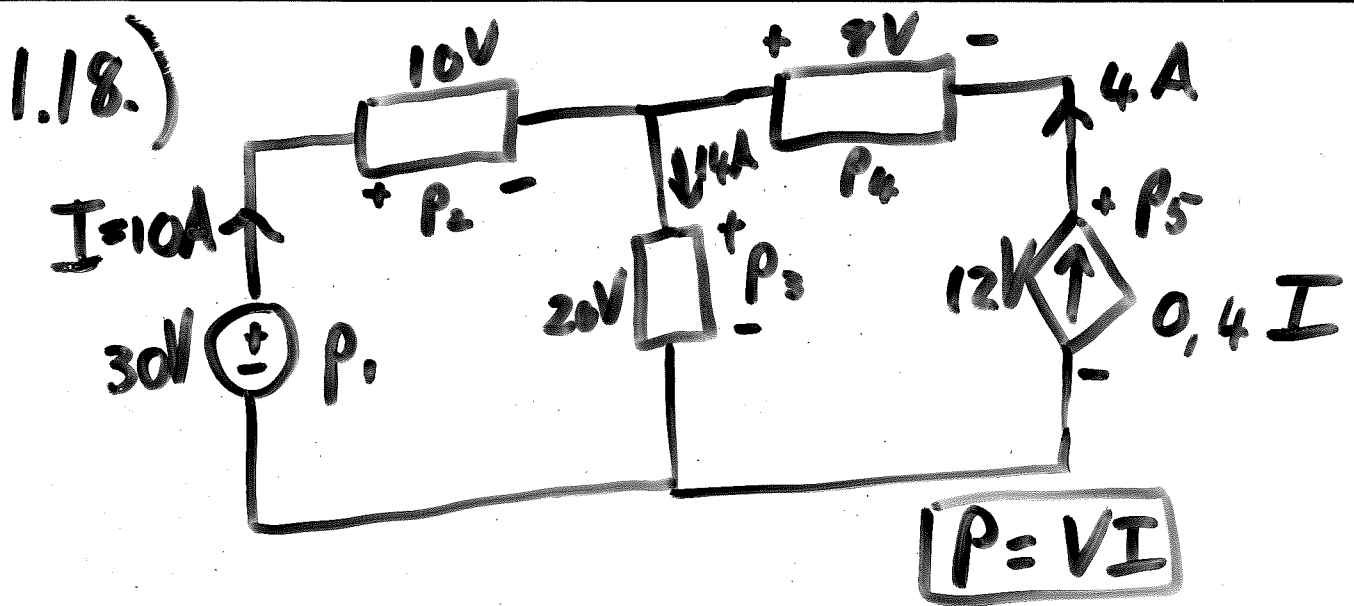
$$\begin{aligned} p_1 &= 60 \text{ W} \\ p_2 &= ? \\ p_3 &= -14.5 \text{ W} \\ p_4 &= 75 \text{ W} \end{aligned}$$

$$\therefore \sum P = 0$$

$$p_1 + p_2 + p_3 + p_4 = 0$$

$$60 + p_2 + (-14.5) + 75 = 0$$

$$\underline{p_2 = 10 \text{ W}}$$



$$P_1 = (30)(-10) = \underline{-300W} \rightarrow$$

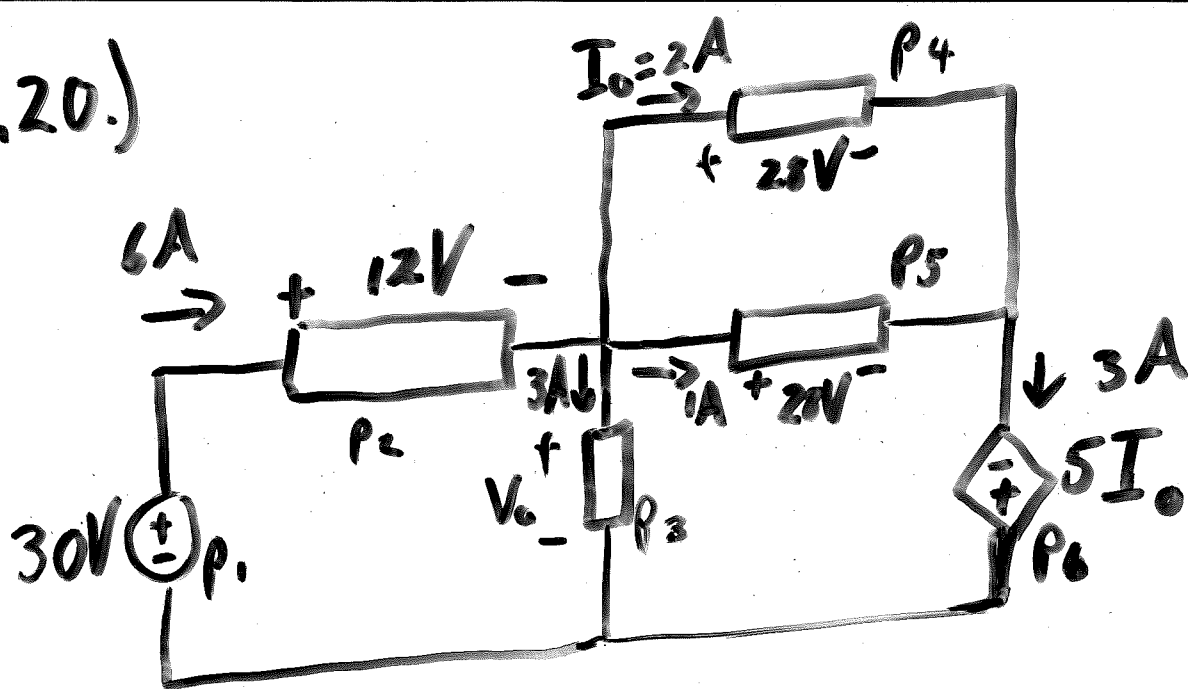
$$P_2 = (10)(10) = \underline{100W} \rightarrow$$

$$P_3 = (20)(14) = \underline{280W} \rightarrow$$

$$P_4 = (8)(-4) = \underline{-32W} \rightarrow$$

$$P_5 = (12)(-4) = \underline{-48W} \rightarrow$$

1.20.)



① $\sum P = 0$

$$(30)(-6) + (12)(6) + V_0(3) + (28)(2) + (2)(1) + (5 \times 2)(-3) = 0$$

$V_0 = 18V$

② $P_1 = (30)(-6) = \underline{-180W}$

$P_2 = (12)(6) = \underline{72W}$

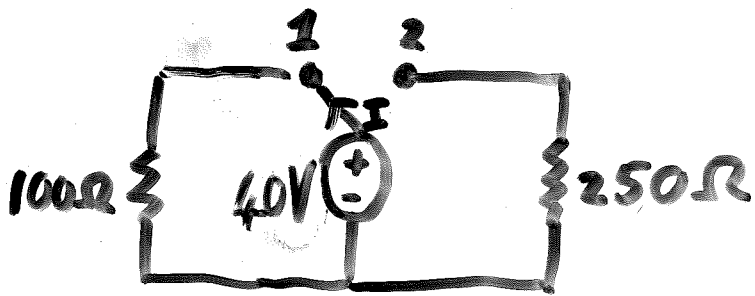
$P_3 = (18)(3) = \underline{54W}$

$P_4 = (28)(2) = \underline{56W}$

$P_5 = (2)(1) = \underline{2W}$

$P_6 = (5 \times 2)(-3) = \underline{-30W}$

2.4.)



a.) Position 1:

$$R = \frac{V}{I}$$

$$100 = \frac{40}{I}$$

$$I = 0,4 \text{ A} \rightarrow$$

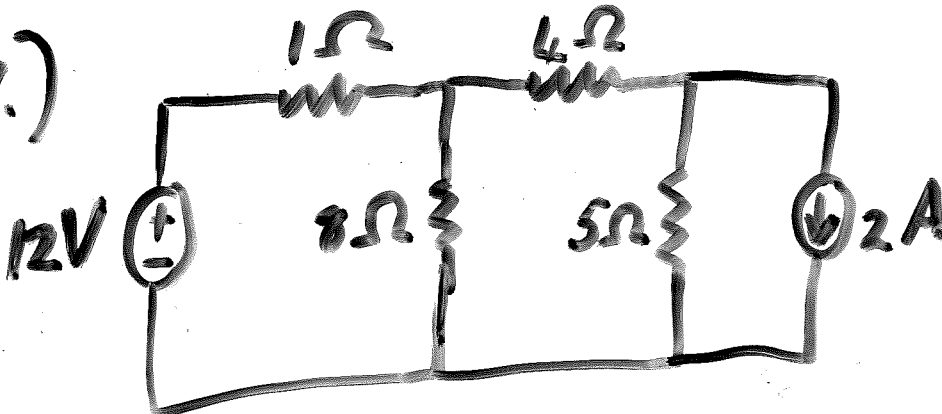
b.) Position 2:

$$R = \frac{V}{I}$$

$$250 = \frac{40}{I}$$

$$I = 0,16 \text{ A} \rightarrow$$

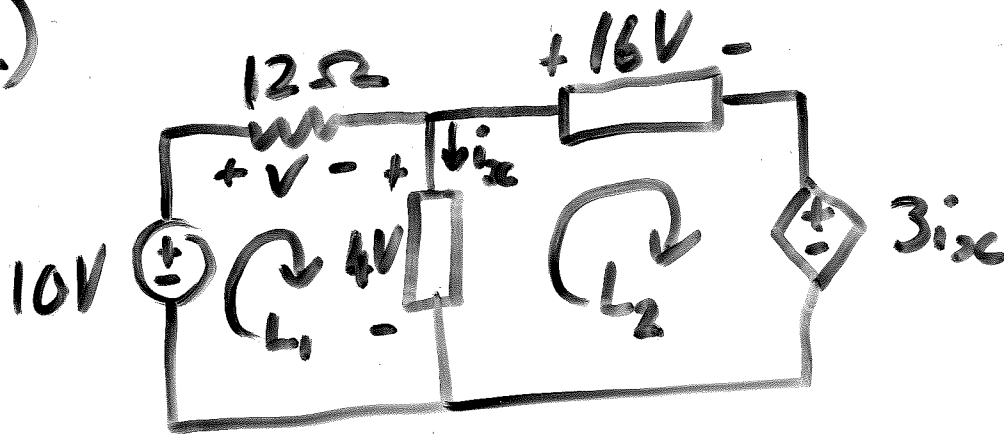
2.7.)



Nodes = 4 →

Branches = 6 →

2.15.)



KVL:

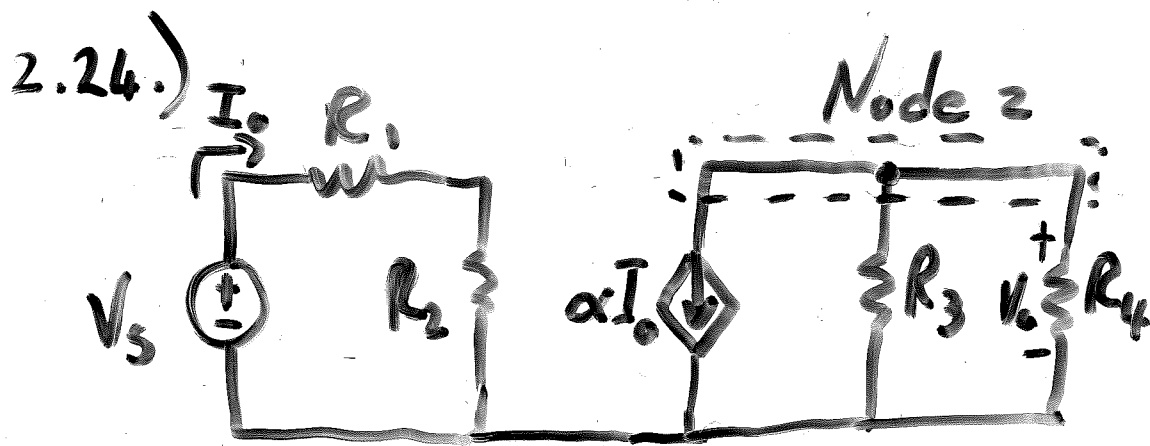
$$L_1 \Rightarrow -10 + V + 4 = 0$$

$$\underline{V = 6V}$$

$$L_2 \Rightarrow -4 + 16 + 3i_x = 0$$

$$3i_x = -12$$

$$\underline{i_x = -4A}$$



$$\alpha = ? \quad \text{if} \quad \left| \frac{V_0}{V_s} \right| = 10$$

$$\textcircled{1} \quad R = \frac{V}{I} \Rightarrow (R_1 + R_2) = \frac{V_s}{I_0}$$

$$\Rightarrow I_0 = \frac{V_s}{(R_1 + R_2)}$$

$$\textcircled{2} \quad \underline{\text{KCL at node 2:}} \quad R = R_1 = R_2 = R_3 = R_4$$

$$\alpha I_0 + \frac{V_0}{R_3} + \frac{V_0}{R_4} = 0$$

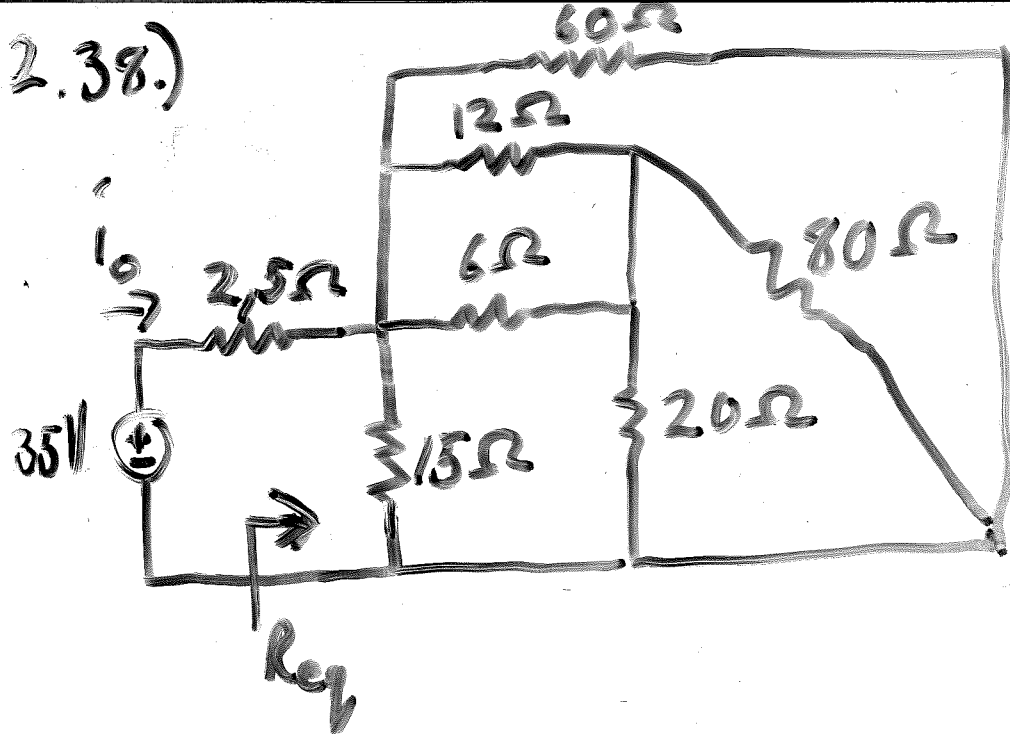
$$\frac{\alpha V_s}{2R} + \frac{2V_0}{R} = 0$$

$$\frac{2V_0}{R} = -\frac{\alpha V_s}{2R}$$

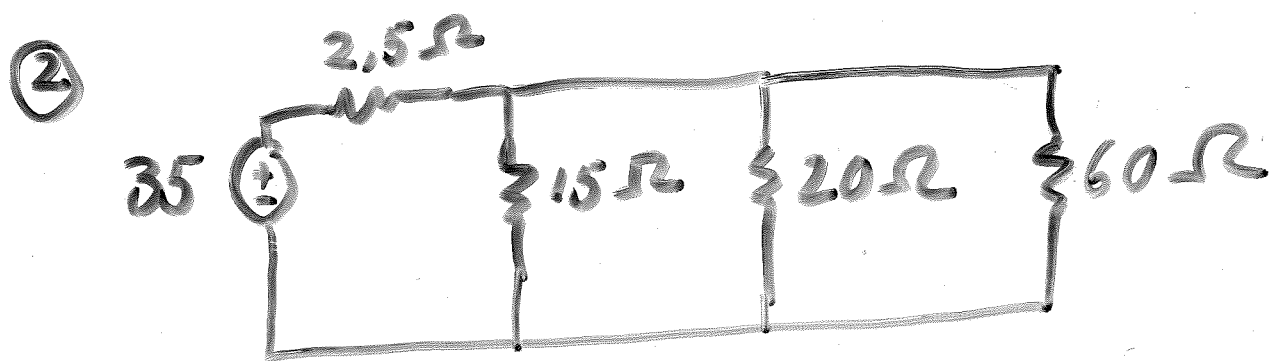
$$\frac{V_0}{V_s} = \frac{\alpha R}{4R}$$

$$10 = \frac{\alpha}{4} \Rightarrow \underline{\underline{\alpha = 40}}$$

2.38.)



$$\textcircled{1} \quad (12/6) + (20/80) = 4 + 16 = \underline{20\Omega}$$



$$\textcircled{3} \quad R_{eq} = 2,5 + (15 \parallel (20 \parallel 60))$$

$$R_{eq} = 2,5 + 7,5 = \underline{10\Omega}$$

$$\textcircled{4} \quad i_o = \frac{35}{R_{eq}} = \frac{35}{10} = \underline{3,5A}$$

Problems worked out during the tutorial session on Friday (29 July 2016).

1.19 $I = -6\text{A}$

$P_1 = -150\text{W}$ (Current Source)

$P_1 = 60\text{W}$

$P_1 = 54\text{W}$

$P_1 = 36\text{W}$

2.12 $V_1 = 70\text{V}$

$V_2 = 10\text{V}$

$V_3 = 60\text{V}$

2.23 $V_x = -100\text{V}$

$P_6 = 960\text{W}$ (Power absorbed by 60 ohm resistor)

2.30 $R_{eq} = 265\text{ Ohm}$

2.35 $I_o = 800\text{mA}$

$V_o = 32\text{V}$

2.45 a.) $R_{eq} = 59.8\text{ Ohm}$

b.) $R_{eq} = 32.5\text{ Ohm}$